

**Spatial Clustering of Suicide Mortality of Older Adult and Its Socioeconomic Determinants
in Illinois (2015–2023)**

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Abstract

Suicide among older adults is a rising public health issue in the United States, but limited studies have examined their spatial distribution at the sub-state level. This study investigated whether the suicide mortality rate of adults aged 65 and over in Illinois showed spatial clustering, and how socio-economic factors were related to the observed patterns. An spatial analysis was carried out using county-level suicide data aggregated through the 2020-2023 Illinois violent death reporting system and socio-economic indicators surveyed by the American community Survey. Choropleth mapping, Global Moran's I, Getis-Ord G_i^* and Local Moran's I are used to identify spatial clustering, followed by ordinary least squares and geographically weighted regression to check spatial variable relationships. The results show that there is a significant spatial autocorrelation in the suicide mortality rate of the elderly. High-risk clusters are concentrated in northeastern Illinois, and cold spots are concentrated in the south and western regions. Geographically weighted regression reveals the spatial heterogeneity of the effects of population density, living alone and work intensity. These findings highlight the importance of place-based approaches in understanding late-life suicide risk and support the use of spatial analysis to inform targeted prevention strategies for older adults.

Keywords: Older adult suicide; Spatial autocorrelation; GWR; Socio-Economic Variable

Introduction

In the United States, the suicide rate of adults aged 65 and above has risen from 15 deaths per 100,000 people in 2009 to 17.7 deaths in 2023 (America's Health Rankings). Illinois Violent Death Reporting System (IVDRS) shows an increase in suicide deaths in Illinois between 2015 and 2023. In 2023, adults aged 65 and older accounted for approximately 19.1% of all suicide deaths in Illinois, despite comprising only 17.6% of the state's total population. This trend indicates that the elderly is increasingly prone to suicide, and it's necessary to explore the socio-economic factor behind it.

An exploratory analysis by Ryan et al. (2022), which identified significant spatial clustering of suicide and related mental health outcomes among adults aged 65 years and older in North Carolina, especially in low-income and rural communities, using violent death reporting system data and spatial scan statistics. However, Illinois lacks a statewide spatial analysis focused on older adult suicide.

Research Questions / Hypothesis

Are suicide deaths among adults aged 65 years and older in Illinois spatially clustered?

Hypothesis: Suicide mortality in Illinois exhibits significant spatial clustering.

What county-level socioeconomic and demographic factors are associated with suicide risk?

Hypothesis: Counties with higher levels of poverty among older adults, greater social isolation, lower educational attainment, and more rural settlement patterns.

This study aims to identify spatial clustering of suicide mortality to examine geographic disparities using spatial autocorrelation, hotspot analysis. Additionally, it assesses the associations between socioeconomic factors and suicide rate through OLS and GWR models.

Materials and Methods

Data Sources and Study Setting:

The study population includes adults aged 65 years and older residing in 102 Illinois counties between 2020 and 2023. County-level aggregation was selected to align with available mortality data, socioeconomic indicators, and public health planning jurisdictions.

County-level suicide data for adults aged 65 years and older were obtained from the IVDRS through the IDPH online dashboard. Suicide deaths were aggregated into total counts and rates across the 2020–2023 period to mitigate the effects of small cell counts and data suppression implemented for privacy protection when fewer than five deaths occur.

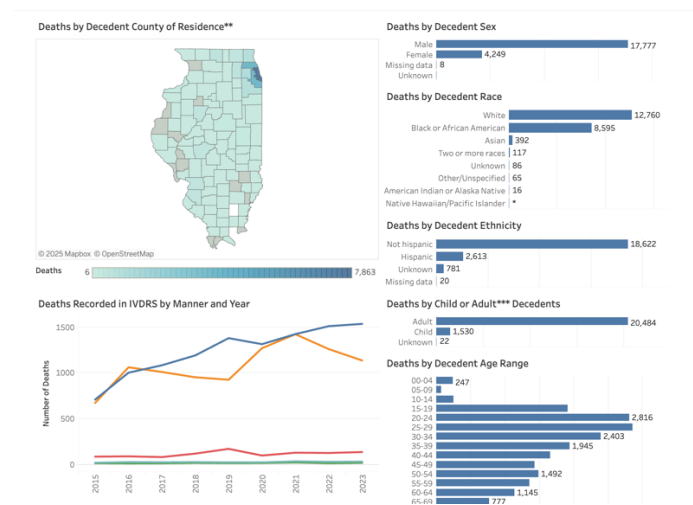


Figure 1. IDPH online dashboard

County-level socioeconomic and demographic variables were obtained from the American Community Survey (ACS) 5-year estimates. Guided by prior suicide research and systematic reviews identifying factors for suicide (Ryan et al., 2022; Qian et al., 2023), this exploratory analysis selected variables reflecting economic disadvantage, social isolation, demographic composition, and settlement context.

Variables: population density as a proxy for urban–rural context, the percentage of adults aged 65 and older living in poverty, the proportion of nonfamily households living alone, the percentage of residents working 35 or more hours per week, the proportion of veterans aged 65 and older, educational attainment, and the percentage of non-Hispanic White residents.

All socioeconomic variables were spatially joined to county boundary polygons and standardized as percentages or rates to facilitate comparison across counties.

Statistical Analysis

Spatial statistical analyses were conducted to examine the geographic distribution, clustering patterns, and socioeconomic correlates of suicide mortality among adults aged 65 years and older in Illinois, following exploratory spatial analysis approaches commonly used in suicide hotspot detection studies (e.g., Bharti & Minz, 2022).

Project the data to UTM Zone 16N to fit the study area. Choropleth mapping of 2020-2023 suicide counts and rates were mapped to visualize county-level patterns of older adult suicide mortality in Illinois. Global spatial autocorrelation was then assessed using Global Moran's I, with incremental spatial autocorrelation employed to identify the distance at which clustering was most significant. To examine localized patterns, Getis–Ord G_i^* hotspot analysis and Anselin Local Moran's I were applied to identify statistically significant clusters, cold spots, and spatial outliers. Finally, relationships between older adult suicide mortality and county-level socioeconomic and demographic factors were evaluated through exploratory regression, ordinary least squares (OLS), and geographically weighted regression (GWR), allowing for the assessment of both global associations and spatially varying relationships.

Results

Choropleth mapping of county-level suicide mortality among adults aged 65 and older revealed clear geographic variation across Illinois during the 2020–2023 period (Figures 2 and 3). Higher suicide counts were concentrated in more populous counties, while population-adjusted suicide rates highlighted elevated risk in several northern and central counties.

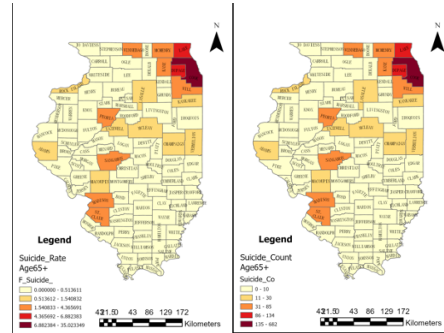


Figure 2, 3: Choropleth mapping of 2020-2023 suicide counts and rates

Incremental spatial autocorrelation analysis was conducted. The analysis identified a clear peak in spatial autocorrelation at distance 56.8 km, corresponding to the maximum z-score = 9.32 (Figure 4). This distance was subsequently used to inform the spatial weighting structure for local cluster analyses. The Global Moran's I statistic was 0.2895, with a z-score of 10.09 and a p-value < 0.001 (Figure 5), indicating a strong positive spatial autocorrelation.

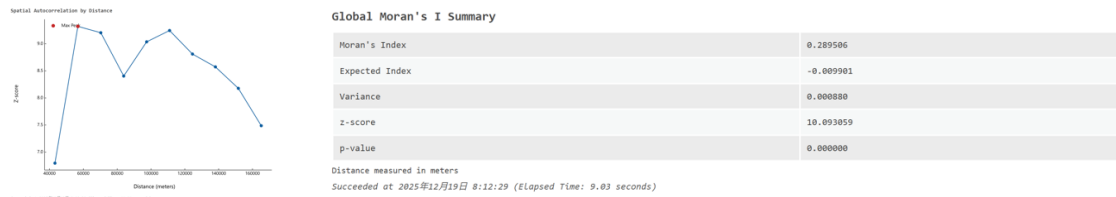


Figure 4, 5: Spatial autocorrelation by distance & Global Moran

Getis–Ord Gi* hotspot analysis identified statistically significant high-risk clusters in counties adjacent to the Chicago metropolitan region (Figure 6). Anselin Local Moran's I revealed multiple high–high and low–low clusters, indicating areas where counties shared similarly elevated or reduced suicide rates (Figure 7).

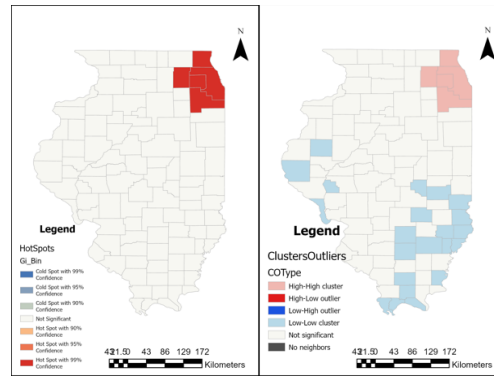


Figure 6, 7: Hot Spot & Cluster and Outlier Analysis

Based on a combination of adjusted R^2 , low multicollinearity, and theoretical relevance to late-life suicide, the model including population density, living alone, and work intensity (≥ 35 hours per week) was selected for geographically weighted regression (GWR) analysis (Figure 8). Despite not passing all global diagnostic thresholds, it offered the strongest overall explanatory performance among candidate models.

Choose 3 of 7 Summary

Highest Adjusted R-Squared Results

AdjR2	AICc	JB	K(BP)	VIF	SA	Model
0.91	311.70	0.00	0.00	1.11	0.55	-WORK35** +LIVINGALONE** +POPDENSITY***
0.91	314.10	0.00	0.00	1.20	0.93	-WORK35** +IMPOVERTYES +POPDENSITY***
0.91	315.07	0.00	0.00	1.61	0.65	+F_WHITE_A -WORK35** +POPDENSITY***

Passing Models

AdjR2	AICc	JB	K(BP)	VIF	SA	Model
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Figure 8: -WORK35** +LIVINGALONE** +POPDENSITY***

GWR reveals the substantial spatial heterogeneity of the relationship between the suicide mortality rate of the elderly and the selected county-level predictors (Figure 9-12). Compared with the global OLS model, the spatial distribution of standardized residuals shows a reduction in clustering. The coefficient chart further shows that the intensity and direction of association throughout Illinois vary greatly. Population density is generally negatively correlated with the suicide mortality rate of the elderly. The impact of living alone has always been positive, but spatially unbalanced, with the strongest links concentrated in southern and southeastern Illinois.

In contrast, the work intensity coefficient (≥ 35 hours per week) shows significant regional differences.

Overall, these results show that the relationship between the socio-economic background and the suicide mortality rate of the elderly is spatially non-fixed, highlighting the importance of a local, location-based approach to understanding and addressing the risk of suicide in old age.

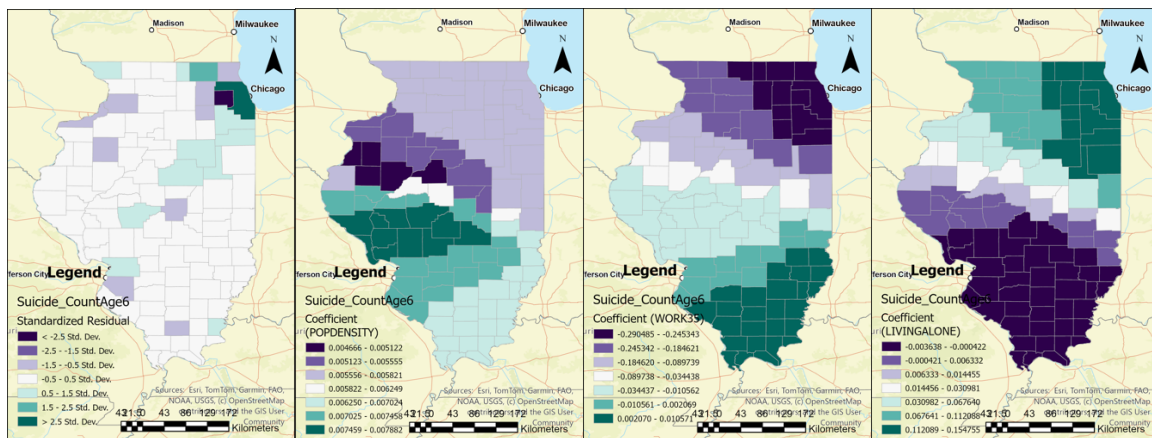


Figure 9, 10, 11, 12: GWR Residuals and Coefficients

Discussion / Conclusions

This study provides evidence that suicide mortality among adults aged 65 and older in Illinois is spatially clustered. Global and local spatial autocorrelation analyses identified hot spots in northeastern Illinois, counties adjacent to the Chicago metropolitan area, while cold spots were more common in southern and western parts of the state.

Exploratory regression and geographically weighted regression further revealed that the associations between suicide mortality and socioeconomic factors are correlated. Higher suicide risk was generally associated with lower population density, supporting vulnerability in rural areas. Living alone showed a consistently positive association with suicide mortality, highlighting the role of social isolation among older adults. In contrast, the effect of work intensity varied geographically, suggesting regional differences.

Several limitations should be noted. First, the use of county-level data may cover up important differences within the county; future research will benefit from more refined geographical units, such as census areas or postal codes. Secondly, due to data suppression and missing values related to privacy, space-time analysis is not feasible. Suicide counts from 2018–2023 were therefore aggregated to ensure data stability. Access to non-suppressed or restricted-use data would enable future space–time cluster detection and improve understanding of temporal dynamics in late-life suicide risk.

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